

Yichuan Liu

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Education

Georgia Institute of Technology

Bachelor of Science in Electrical Engineering, GPA 4.00

August 2023 – Present

Expected Graduation: Dec 2027

Coursework: Computer Architecture, Machine Learning, Signal Processing, Programming for HW/SW Systems, Signals and Systems, Microelectronics, Digital Design, Circuit Analysis

Awards: \$10,000 Undergraduate Research Grant; NT\$7000 Synopsys SNUG Taiwan Presentation Award

Professional Experience

GPU Performance Engineering Intern | NVIDIA, Santa Clara, CA

August 2026 – Present

- Benchmark and optimize deep-learning and HPC workloads on GPUs for latency, throughput, and power efficiency

Machine Learning Research Intern (EDA) | TENSTORRENT, Austin, TX

May 2026 – August 2026

- Developed WaveformQA, the first benchmark evaluating frontier LLM temporal reasoning on RISC-V waveforms
- Applied benchmark insights into internal agentic EDA workflows and increased waveform debug accuracy by 37-53%
- Published this research as a **first-author [C1]** and presented the findings at Stanford University

CPU Design Verification Intern | TENSTORRENT, Austin, TX

January 2026 – May 2026

- Deployed a novel Reinforcement Learning framework into verification flows for RISC-V AI Accelerators using Bazel
- Integrated solution directly into Tenstorrent's active production pipeline, reducing regression runtime by 57%
- Led a **first-author** publication **[C2]** with Synopsys, and received NT\$7000 Synopsys Presentation Award

Software Engineering Intern | BMW, Munich, BY

May 2024 – August 2024

- Developed a data visualization testing application for child presence detection for 10000+ Ethernet and CAN signals
- Solution adopted by 3 departments (400+ users) and launched on BMW's server
- Led a team of 5 interns, coordinating development tasks and ensuring successful project delivery

Conference Publications

- [C1] Yichuan Liu***, D. Cummings*, et al. *WaveformQA: Benchmarking LLM Temporal Reasoning on Digital Waveforms*. **IEEE ICLAD'26**, presented at Stanford University (long paper, oral)
- [C2] Yichuan Liu**, R. Gehani, et al. *AI-Driven Coverage Optimization for RISC-V MMU Verification*. **SNUG Taiwan** (electronics industry's largest user conference, oral)

Journal Publications

- [J1] Yichuan Liu**, L. Graber, et al. *Early Prediction of Metal Oxide Varistor Lifetime in DC Circuit Breakers using Feature Engineering and Machine Learning*. **IEEE Transactions on Industry Applications** (Under Review, 2025)
- [J2]** Yang Liu, K. Chuong, **Yichuan Liu**, L. Graber, et al. *Degradation, Lifetime, and Failure Modes of Metal Oxide Varistors in Low-Voltage DC Circuit Breaker Applications*. **IEEE Transactions on Power Delivery**

Research Experience

AI Accelerator Researcher | SYNERGY Lab Georgia Tech

August 2025 – January 2026

- Developed neuro-symbolic AI models that integrate LLMs with PDDL planners for robotics task planning
- Benchmarked system performance on CPU/GPU architectures to identify bottlenecks, achieving a 23% reduction in latency and a 5% throughput increase

Machine Learning Researcher | PDL Lab Georgia Tech

January 2024 – January 2026

- Developed first generalizable machine learning models (LSTM, biGRU, ElasticNet) in PyTorch to predict metal oxide varistor lifetime with 94.6% accuracy using only 10.5% of early degradation data; led to **first-author** journal **[J1]**
- Analyzed 23 billion data points, correlating material degradation with electrical changes for DC MOV Circuit Breaker using regression models and signal processing; led to co-author journal **[J2]**

Skills

Tools: Synopsys VCS & Verdi, Bazel, Linux, Docker, Git/GitLab, Intel Quartus, GTKWave, Icarus Verilog, Modelsim

Programming: Python, CUDA, Bash, Assembly, C++, MATLAB, SQL

ML & Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Hardware: RTL Design, SystemVerilog, FPGA, VLSI, Microprocessors, Logic Analyzers, Oscilloscopes, Multimeters